

# MICHAEL JAYASURIYA

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## EDUCATION

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### Drexel University College of Medicine

Philadelphia, PA

*Doctor of Medicine (M.D.) Candidate*

Expected May 2028

- Completed third year; research fellowship year 2026 – 2027. Applying to Neurological Surgery residency (2028 Match)

### University of California, Berkeley

Berkeley, CA

*B.A. Computer Science (Honors), GPA: 3.8*

2018 – 2022

## EXPERIENCE

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### Section of Computational Neurosurgery, University of Chicago

Chicago, IL

*Research Fellow (PI: Mohamad Bydon, M.D.)*

2025 – Present

- Developing deep learning pipelines for automated spine imaging phenotyping (CT/MRI vertebral segmentation, vertebral bone quality, spinopelvic parameters) for the multicenter QOD Spine CORE registry
- Fine-tuning privacy-preserving large language models (Qwen3, QLoRA) for structured information extraction from spine operative notes
- Supporting external validation of the automated vertebral bone quality (VBQ) algorithm with the University of Florida

### Department of Neurosurgery, Mayo Clinic & Neuroinnovations, LLC

Rochester, MN

*Research Assistant & AI Software Engineer (PI: Mohamad Bydon, M.D.)*

May 2023 – 2025

- Applied deep learning to spine imaging: developed models for automated osteoporosis screening from CT/MRI, SINS calculation from CT, and compression fracture grading from x-ray
- Built causal ML pipelines (uplift modeling, AIPW) to predict individualized outcomes for robotic spine surgery and early discharge using NSQIP data
- Helped achieve ISO certification for Spine Studio, an AI-powered spine surgery planning software product at Neuroinnovations, LLC
- Contributed to NIH STTR and NVIDIA academic researcher grant applications

### Google's Self-Driving Car Project

Mountain View, CA

*Software Engineer, Waymo (Alphabet subsidiary)*

Jul 2022 – Aug 2023

- Designed and trained an image captioning neural network to accelerate annotation of vehicle collision events
- Reduced build times by 50% and eliminated reliance on deprecated components

### Verily (Google Life Sciences)

South San Francisco, CA

*Software Engineering Intern*

May 2021 – Aug 2021

- Developed an Android voice recording app for opioid rehabilitation patients, still in use at the clinic OneFifteen in Dayton, Ohio
- Built server-side GraphQL in TypeScript for speech-to-text transcription and sentiment analysis

### Big Data in Radiology Lab, UCSF

San Francisco, CA

*Undergraduate Researcher*

Dec 2021 – May 2022

- Implemented an end-to-end content-based image retrieval system for chest x-ray and CT scan data using a novel vision transformer in PyTorch

### Yosef Lab, Center for Computational Biology, UC Berkeley

Berkeley, CA

*Undergraduate Researcher*

Dec 2020 – May 2022

- Contributed to scvi-tools (Nature Biotechnology, 2022) and developed novel variational autoencoders for single-cell omics data analysis

## PEER-REVIEWED PUBLICATIONS

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1. Gayoso A, Lopez R, Xing G, Wu K, Jayasuriya M, et al. "scvi-tools: a library for deep probabilistic analysis of single-cell omics data." *Nature Biotechnology*, 2022.
2. Ergen C, Xing G, Xu C, et al. (including Jayasuriya M). "Consensus prediction of cell type labels in single-cell data with popV." *Nature Genetics*, 2024.
3. Feng E, Jayasuriya NM, Nathani KR, Katsos K, Machlab LA, Johnson GW, et al. "Artificial intelligence image analysis for Hounsfield units in preoperative thoracolumbar CT scans: an automated screening for osteoporosis in patients undergoing spine surgery." *Journal of Neurosurgery: Spine*, 2025, 43(1), 1–8. DOI: 10.3171/2025.1.SPINE24900.
4. Jayasuriya NM, Feng E, Nathani KR, Delawan M, Katsos K, Bhagra O, Freedman BA, Bydon M. "Automated vertebral bone quality score measurement on lumbar MRI using deep learning: Development and validation of an AI algorithm." *Clinical Neurology and Neurosurgery*, 2025, 257, 109094. DOI: 10.1016/j.clineuro.2025.109094.
5. Feng E, Jayasuriya N, Kirby D. "Artificial Intelligence in Orthopedic Surgery: A Practical Overview of Current Applications and Trends." *SurgiColl*, 2025, 3(3). DOI: 10.58616/001c.137606.
6. Nathani KR, Machlab LA, von Kentzinsky HL, Feng E, Jayasuriya NM, et al. "Optimizing neurosurgical operating room schedule: Integrating machine learning models for operative time prediction and schedule design." *Surgical Neurology International*, 2025, 16, 395. DOI: 10.25259/SNI\_626\_2025.

## PREPRINTS

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1. Thakrar A\*, Jayasuriya M\*, Serapio A, Wu X, Davis E, Schroeder J, Vella M, Sohn JH. "Semantic retrieval of similar radiological images using vision transformers." *medRxiv*, 2023. DOI: 10.1101/2023.02.16.23286056. (\*equal contribution)

## CONFERENCE PRESENTATIONS & ABSTRACTS

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### **Spine Summit 2024, Las Vegas, NV**

1. Jayasuriya NM, Feng E, Nathani KR, Katsos K, Ibrahim S, Johnson GW, Freedman B, Bydon M. "AI Image Analysis for Hounsfield Units in Preoperative Spine CT Scans." Oral Presentation, February 2024.
2. Nathani KR, Jayasuriya NM, Feng E, et al. "Automated Measurement of Spinal Radiographic Parameters." Oral Presentation, February 2024.

### **AANS Annual Meeting 2024, Chicago, IL**

3. Feng E, Jayasuriya NM, Nathani KR, Ibrahim S, Katsos K, Johnson GW, Freedman B, Bydon M. "AI Image Analysis for Hounsfield Units in Preoperative Spine CT Scans: An Automated Screening for Osteoporosis in Patients Undergoing Spine Surgery." Oral Presentation, May 2024.

### **CNS Annual Meeting 2024, Houston, TX**

4. Jayasuriya N, Feng E, Nathani KR, Katsos K, Freedman B, Bydon M. "AI Image Analysis for Extracting Vertebral Bone Quality Score from Preoperative MRI: An Automated Screening for Osteoporosis in Patients Undergoing Spine Surgery." Rapid Fire Oral Presentation, September 2024.

### **Spine Summit 2025, Tampa, FL**

5. Feng E, Jayasuriya NM, et al. "Deep Learning Prediction of DEXA T-Score on MRI Outperforms Vertebral Bone Quality Score in Osteoporosis Detection." Oral Presentation, February 2025.
6. Jayasuriya NM, Feng E, et al. "Automated Assessment of Spondylolisthesis Severity on Lumbar Spine X-Rays Using Artificial Intelligence." Oral Presentation, February 2025.

### **AANS Annual Meeting 2025, Boston, MA**

7. Jayasuriya NM, Feng E, et al. "AI-Driven Approach for Automated Detection and Grading of Compression Fractures on Lumbar Spine X-Rays." Oral Presentation, April 2025.

### **ISASS Annual Conference 2025, Miami Beach, FL**

8. Willis KM, Feng E, Jayasuriya NM, Nathani K, Ryan J, Bydon M. "Fine Tuning a Large Language Model to Extract Thoracolumbar Injury Classification Score." Podium Presentation, April 2025.

### **AAOS Annual Meeting 2025**

9. Jayasuriya N, Feng E, Goswani K, Ramtin S, Ilyas AM. "An Artificial Intelligence Tool to Prevent Wrong-Side Hand Surgery: A Feasibility Study." 2025.

### **AAHS Annual Meeting 2025**

10. Jayasuriya N, Feng E, Goswani K, Ramtin S, Ilyas AM. "Deep Learning for Pediatric Wrist Fracture Detection." 2025.

### **EOA Annual Meeting 2025**

11. Jayasuriya N, Feng E, Goswani K, Ramtin S, Ilyas AM. "Deep Learning for Detection of Cervical Fractures." 2025.

### **Spine Summit 2026, Phoenix, AZ**

12. Jayasuriya NM, Feng E, et al. "Are Multimodal Large Language Models Superior to Text-Only Natural Language Processing Models for Automated Information Extraction from Spine Operative Notes? A QOD Spine CORE Study." Oral Presentation, March 2026.
13. Matache I, et al. (including Jayasuriya NM). "Can Large Language Models Accurately Extract Data from Operative Notes on Grade II Spondylolisthesis? A QOD Spine CORE Study." Oral Presentation, March 2026.

### **AANS Annual Meeting 2026, San Antonio, TX**

14. Feng E, Jayasuriya NM, et al. "Estimating the Safety of Early Discharge Using Uplift Modeling: A Causal Machine Learning Analysis of the NSQIP Database." Oral Presentation, May 2026.

### **CNS Annual Meeting 2026, Washington, DC**

15. Jayasuriya NM (presenting author), et al. "An Interpretable Risk Calculator for Glioblastoma Survival Combining Human-Readable MRI Features, MGMT Methylation, and Clinical Variables." Poster Presentation, November 2026. (Accepted)

## **MEDICAL EDUCATION PROJECTS**

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### **PIAT Learn (Patient Interaction Analysis Tool), Drexel University College of Medicine** 2025 – Present

- Co-developed with Emily Feng (mentor: Emily Spengler, M.D.) a voice-based AI standardized patient built on Meta Llama that gives students immediate feedback on spoken grade level, jargon, checks for understanding, and conversational turn-taking; piloted with pediatric residents at St. Christopher's Hospital for Children

## **IN THE PRESS**

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1. "State of the Science: AI in Medical Education." Drexel University College of Medicine Alumni Magazine, Spring 2026. Featured for PIAT Learn.